Roberto Riaño Hidalgo

PhD Researcher · Security of Generative & Neural AI Systems

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SUMMARY

PhD researcher in the security of generative and neural AI systems at Ikerlan Research Centre and Radboud University. I design backdoor and poisoning attacks that expose how models fail, from the first physical-world backdoor on spiking neural networks (ACSAC 2025) to supply-chain backdoors that hijack downstream control through pretrained world models (under review).

PUBLICATIONS & PREPRINTS

[1] R. **Riaño**, G. Abad, S. Picek, A. Urbieta. "Flashy Backdoor: Real-world Environment Backdoor Attack on SNNs with DVS Cameras." *IEEE Annual Computer Security Applications Conference (ACSAC)*, 2025, pp. 986–1002. doi:10.1109/ACSAC67867.2025.00081 · code
Presented at ACSAC 2025, Honolulu, HI.

[2] G. Abad, M. Krček, S. Koffas, B. Tajalli, M. Arazzi, R. **Riaño**, X. Xu, Z. Liu, et al. "SoK: The Last Line of Defense — On Backdoor Defense Evaluation." *arXiv preprint* arXiv:2511.13143, 2025.

[3] R. **Riaño**, G. Abad, S. Picek, A. Urbieta. "When the World Lies: Backdoor Attacks on Latent World Models for Downstream Control." *Under review, IEEE S&P 2026*.

EDUCATION

PhD — Security of Generative AI · Ikerlan Research Centre & Radboud University — 2024 – Present (expected 2028)

MSc — Computational Engineering & Intelligent Systems · UPV/EHU · Grade 9.13/10 — 2023 – 2024

BSc — Computer Engineering (Computer Science specialization) · UPV/EHU — 2019 – 2023

RESEARCH & WORK EXPERIENCE

Doctoral Researcher — Security of Generative AI · *Ikerlan Research Centre & Radboud University* — Sep 2024 – Present

- Researching the security of generative and neural AI systems, including backdoor and poisoning threats to world models (advisors: S. Picek, G. Abad, A. Urbieta).

Master's Thesis — Real-World Environment Backdoor Attack on SNNs · *Ikerlan Research Centre* — Feb – Jul 2024

- Conducted the first study of backdoor attacks on Spiking Neural Networks (SNNs) in real-world physical environments using event-based Dynamic Vision Sensor cameras.
- Designed three novel attack methods (Framed, Strobing, Flashy), reaching up to 100% attack success rate with negligible clean-accuracy loss across three neuromorphic benchmarks.
- Built and released an expanded poisoned DVS128-Gesture dataset for defense evaluation; adapted and evaluated four SOTA defenses (Fine-Pruning, STRIP, Februus, BaDExpert). **Published at IEEE ACSAC 2025.**

Research Intern — Vulnerabilities & Defenses in AI Models · *Ikerlan Research Centre* — Oct 2023 – Feb 2024

- Surveyed adversarial and backdoor attack literature and the corresponding defense and detection techniques; identified under-explored problems in AI security.
- Prototyped adversarial examples and backdoor attacks to characterize model vulnerabilities.

Bachelor's Thesis — Robustness of Autonomous-Driving Systems · *Ikerlan Research Centre* Feb – Jul 2023

- Reviewed the state of the art in DNN-based image detection and adversarial attacks for autonomous driving.
- Integrated a CNN-based traffic-sign detection system and built and evaluated multiple adversarial attacks against it.

Summer Intern — Post-Quantum Cryptography · *Ikerlan Research Centre* Jun – Sep 2022

- Analyzed NIST post-quantum cryptographic algorithms; built proof-of-concept pilots and benchmarked quantum-resistant cryptography for industrial IoT (IIoT) environments.

TECHNICAL SKILLS

Languages: Python (primary), C / C++, C#, Java, R, SQL

ML & tools: PyTorch, Hugging Face, NumPy / Pandas, Matplotlib / Seaborn, Git, Linux

Research areas: AI security · backdoor & poisoning attacks · world models · spiking neural networks & neuromorphic data · generative-model security · adversarial ML

ACADEMIC SERVICE

Reviewer: IEEE Transactions on Information Forensics and Security (T-IFS, 2 reviews); IEEE Transactions on Emerging Topics in Computing (TETC, 2 reviews)

GRANTS & FUNDED PROJECTS

Contributed to a successfully funded European research proposal in generative AI security (announcement pending; details available on request).